

M21 2026 July 2026 N400 Component

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1 Setup

Load libraries

Set emmeans options

```
emm_options(  
  lmer.df = "kenward-roger",  
  pbkrtest.limit = 10000,  
  disable.pbkrtest = FALSE  
)
```

Set ggplot parameters

```
# Color palette  
my_palette_1 <- c(  
  "#E31A1C", # red  
  "#1F78B4" # blue  
)  
  
my_palette_2 <- c(  
  "#0072B2", # dark blue  
  "#E69F00" # orange/gold  
)  
my_linetypes_1 <- c("dashed", "solid")  
  
# Theme  
my_theme <- theme_classic(base_size = 8) +  
  theme(  
    legend.position = "bottom",  
    legend.direction = "horizontal",  
    legend.box = "horizontal"  
  )  
  
# Common guides  
my_guides <- guides(  
  colour = guide_legend(nrow = 1),  
  fill = guide_legend(nrow = 1)  
)
```

Define 2-way interaction table function

Function use for Complexity $\times$ LP table:

```
emm_complexity_LP_N400 <- emmeans(  
  m_nw_N400_fs,  
  ~ complexity | LP,  
  at = list(LP = c(-1, 0, 1))  
)  
  
compact_emm_table(  
  emm_complexity_LP_N400,  
  by_var = LP,  
  condition_prefix = "LP"  
)
```

for Family Size $\times$ LP table:

```
emm_FS_LP_N400 <- emmeans(  
  m_nw_N400_fs,  
  ~ family_size | LP,  
  at = list(LP = c(-1, 0, 1))  
)  
  
compact_emm_table(  
  emm_FS_LP_N400,  
  by_var = LP,  
  condition_prefix = "LP"  
)
```

For 3-way use this function like this:

```
emm_3way <- emmeans(  
  m_nw_N400_fs,  
  ~ complexity * family_size | LP,  
  at = list(LP = c(-1, 0, 1))  
)  
  
compact_3way_table(  
  emm_3way,  
  by_var = LP,  
  condition_prefix = "LP",  
  factor1 = complexity,  
  factor2 = family_size  
)
```

)

2 Load Files

```
# Load data: rt words
rt_w <- read_csv("rt_words_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    Item = factor(target),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") |>
  filter(correct == 1) |>
  mutate(
    SubjID = factor(SubjID),
    ItemID = factor(ItemID),
    complexity = factor(complexity,
      levels = c("SIMP", "COMP"),
      labels = c("Simple", "Complex")),
    logRT = log(response_time),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1),
    zLogBF = as.numeric(scale(LogBF)),
    zLogFS = as.numeric(scale(LogFS))
  )

# Load data: erp_fs words
erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    family_size = factor(family_size,
      levels = c("Small", "Large")),
    complexity = factor(complexity,
      labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>
  mutate(
    SubjID = factor(SubjID),
    chlabel = factor(chlabel),
    base_frequency = factor(base_frequency,
      levels = c("Low", "High")),
```

```

    complexity = factor(complexity,
                        labels = c("Simple", "Complex")),
    time_window = factor(time_window),
    OS = as.numeric(Dim.2),
    LP = as.numeric(Dim.1)
  )
}

range(erp_fs_w$LP)

[[ [1] -2.674523  3.111265
range(erp_fs_w$OS)

[[ [1] -1.975820  1.926921

```

3 N400 WORD DATA: INTERACTION WITH LP (PRIMARY ANALYSES)

3.1 N400 WORD DATA: BASE FREQUENCY and LP as FIXED EFFECTS; OS as COVARIATE

3.1.1 Model

```

contrasts(erp_bf_w$base_frequency) <- c(-0.5, 0.5)

m_w_N400_bf <- lmer(
  value ~ base_frequency * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_bf)

[[ Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
[[ Formula: value ~ base_frequency * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
[[ Data: subset(erp_bf_w, time_window == "N400")
[[ Control: lmerControl(optimizer = "bobyqa")
[[
[[ REML criterion at convergence: 19468.7
[[
[[ Scaled residuals:
[[      Min       1Q   Median       3Q      Max
[[ -11.6094  -0.3522   0.0107   0.3807  10.5439
[[
[[ Random effects:
[[ Groups      Name                Variance Std.Dev.
[[ SubjID:chlabel (Intercept)  23.249    4.822
[[ SubjID       (Intercept)  18.541    4.306
[[ Residual                    9.157    3.026
[[ Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
[[
[[ Fixed effects:
[[              Estimate Std. Error   df t value Pr(>|t|)
[[ (Intercept)    5.617e-01  5.722e-01  5.700e+01  0.982  0.3304
[[ base_frequency1 2.141e-01  1.064e-01  1.618e+03  2.012  0.0444 *
[[ LP             4.677e-02  4.652e-01  5.700e+01  0.101  0.9203
[[ OS             1.290e+00  6.399e-01  5.700e+01  2.016  0.0485 *
[[ base_frequency1:LP 3.444e-01  8.659e-02  1.618e+03  3.977  7.29e-05 ***
[[ ---
[[ Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
[[
[[ Correlation of Fixed Effects:
[[      (Intr) bs_fr1 LP      OS
[[ bas_frqncy1  0.000
[[ LP          0.038  0.000
[[ OS          -0.046  0.000  0.021
[[ bs_frqn1:LP  0.000  0.039  0.000  0.000
[[
[[ # Effect sizes for individual terms
eta_squared(m_w_N400_bf, partial = TRUE)

[[ # Effect Size for ANOVA (Type III)
[[
[[ Parameter          | Eta2 (partial) |      95% CI
[[ -----
[[ base_frequency     |      2.50e-03 | [0.00, 1.00]

```

```

|| LP | 1.77e-04 | [0.00, 1.00]
|| OS | 0.07 | [0.00, 1.00]
|| base_frequency:LP | 9.68e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# Model R²
r2_nakagawa(m_w_N400_bf)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.825
|| Marginal R2: 0.026

```

3.1.2 Means and Contrasts

```

as.tibble(emmeans(m_w_N400_bf, ~ base_frequency)) |> print()

```

```

|| # A tibble: 2 x 6
|| base_frequency emmean SE df lower.CL upper.CL
|| <fct> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Low 0.515 0.574 58.0 -0.634 1.66
|| 2 High 0.712 0.574 58.0 -0.436 1.86

```

```

emm_bf_LP <- emmeans(m_w_N400_bf,
  ~ base_frequency | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)))

```

```

emm_part <- summary(emm_bf_LP) |>
  as.data.frame() |>
  transmute(LP, Effect = paste("EMM:", base_frequency), Estimate = emmean, SE, df,
    lower.CL, upper.CL, t.ratio = NA_real_, p.value = NA_real_)

```

```

contrast_part <- pairs(emm_bf_LP) |>
  summary() |>
  as.data.frame() |>
  transmute(LP, Effect = paste("Contrast:", contrast), Estimate = estimate, SE, df,
    lower.CL = NA_real_, upper.CL = NA_real_, t.ratio, p.value)

```

```

emm_contrast_table_LP_N400_words <- bind_rows(emm_part, contrast_part) |>
  arrange(LP, Effect)|>
  mutate(
    Estimate = round(Estimate, 2),
    SE = round(SE, 2),
    df = round(df, 1),
    lower.CL = round(lower.CL, 2),
    upper.CL = round(upper.CL, 2),
    t.ratio = round(t.ratio, 2),
    p.value = round(p.value, 3)
  )

```

```

as.tibble(emm_contrast_table_LP_N400_words, digits = 3) |> print(n=20)

```

```

|| # A tibble: 15 x 9
|| LP Effect Estimate SE df lower.CL upper.CL t.ratio p.value
|| <dbl> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 -2.5 Contrast: Low - High 0.65 0.24 1618 NA NA 2.72 0.007
|| 2 -2.5 EMM: High 0.18 1.28 58 -2.39 2.74 NA NA
|| 3 -2.5 EMM: Low 0.82 1.28 58 -1.74 3.39 NA NA
|| 4 -1 Contrast: Low - High 0.13 0.13 1618 NA NA 0.97 0.333
|| 5 -1 EMM: High 0.5 0.73 58 -0.95 1.96 NA NA
|| 6 -1 EMM: Low 0.63 0.73 58 -0.82 2.09 NA NA
|| 7 0 Contrast: Low - High -0.21 0.11 1618 NA NA -2.01 0.044
|| 8 0 EMM: High 0.72 0.57 58 -0.43 1.87 NA NA
|| 9 0 EMM: Low 0.51 0.57 58 -0.64 1.66 NA NA
|| 10 1 Contrast: Low - High -0.56 0.14 1618 NA NA -3.99 0
|| 11 1 EMM: High 0.94 0.75 58 -0.57 2.45 NA NA
|| 12 1 EMM: Low 0.38 0.75 58 -1.13 1.89 NA NA
|| 13 3 Contrast: Low - High -1.25 0.28 1618 NA NA -4.38 0
|| 14 3 EMM: High 1.38 1.54 58 -1.69 4.45 NA NA
|| 15 3 EMM: Low 0.13 1.54 58 -2.94 3.21 NA NA

```

3.1.3 Plots

3.2 N400 WORD DATA: FAMILY SIZE and LP as FIXED EFFECTS; OS as COVARIATE

3.2.1 Model

```
contrasts(erp_fs_w$family_size) <- c(-0.5, 0.5)

m_w_N400_fs <- lmer(
  value ~ family_size * LP + OS +
    (1 | SubjID) +
    (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_fs)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * LP + OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15561.8
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9307  -0.3874   0.0081   0.4071   6.2048
||
|| Random effects:
|| Groups          Name                Variance Std.Dev.
|| SubjID:chlabel (Intercept)  5.238      2.289
|| SubjID         (Intercept)  4.575      2.139
|| Residual                                3.304      1.818
|| Number of obs: 3240, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    2.618e-01  2.842e-01  5.700e+01  0.921  0.3608
|| family_size1    4.522e-01  6.392e-02  1.618e+03  7.074 2.23e-12 ***
|| LP              1.231e-03  2.311e-01  5.700e+01  0.005  0.9958
|| OS              6.771e-01  3.179e-01  5.700e+01  2.130  0.0375 *
|| family_size1:LP -7.114e-02  5.201e-02  1.618e+03 -1.368  0.1716
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) fmly_1 LP      OS
|| family_siz1  0.000
|| LP           0.038  0.000
|| OS          -0.046  0.000  0.021
|| fmly_sz1:LP  0.000  0.039  0.000  0.000

# Effect sizes for individual terms
eta_squared(m_w_N400_fs, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter      | Eta2 (partial) |      95% CI
|| -----|-----|-----|
|| family_size    |      0.03 | [0.02, 1.00]
|| LP             | 4.98e-07 | [0.00, 1.00]
|| OS             |      0.07 | [0.00, 1.00]
|| family_size:LP | 1.15e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

# Model R²
r2_nakagawa(m_w_N400_fs)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.756
|| Marginal R2: 0.031
```

3.2.2 Means

```
as.tibble(emmeans(m_w_N400_fs, ~ family_size)) |> print()

|| # A tibble: 2 x 6
```

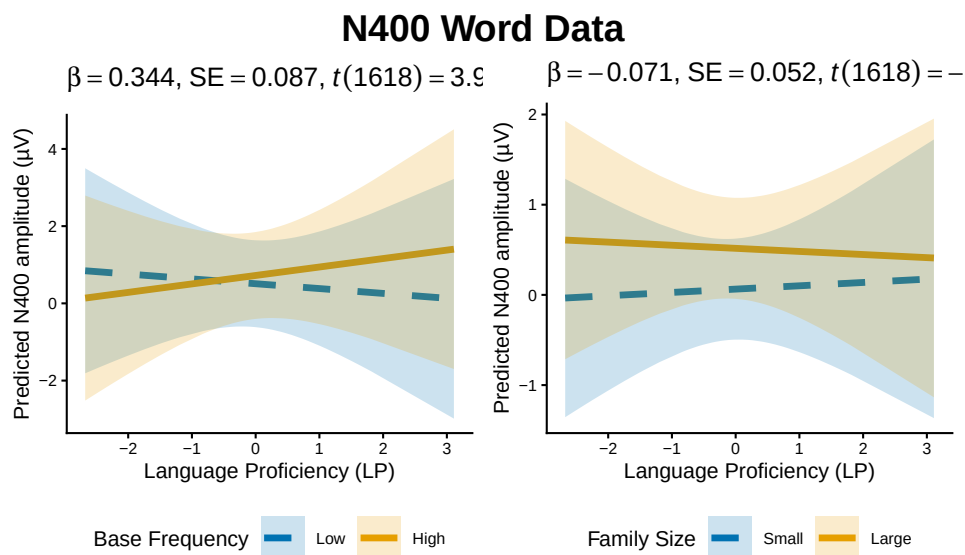
	family_size	emmean	SE	df	lower.CL	upper.CL
	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Small	0.0623	0.286	58.5	-0.509	0.634
2	Large	0.518	0.286	58.5	-0.0535	1.09

3.2.3 Plots (Interaction Not significant)

3.3 N400 WORD DATA: Interaction Plots— BASE FREQUENCY x LP; FAMILY SIZE x LP

```
row1 <- plot_grid(BFxLP_n400, FSxLP_n400, ncol = 2)

plot_grid(
  ggdraw() + draw_label("N400 Word Data", fontface = "bold"),
  row1,
  ncol = 1,
  rel_heights = c(0.08, 1)
)
```



Does the slopes of the lines in the significant interactions, significantly differ from zero?

```
test(entrends(m_w_N400_bf, ~ base_frequency, var = "LP"))
```

	base_frequency	LP.trend	SE	df	t.ratio	p.value
Low	-0.125	0.467	58	-0.268	0.7893	
High	0.219	0.467	58	0.469	0.6411	

Degrees-of-freedom method: kenward-roger

4 N400 WORD DATA: INTERACTION WITH OS (EXPLORATORY ANALYSES)

4.1 N400 WORD DATA: BASE FREQUENCY and OS as FIXED EFFECTS; LP as COVARIATE

```
# BINNED BY BASE FREQUENCY
m_w_N400_bf_OS <- lmer(
  value ~ base_frequency * OS + LP +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_bf_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_bf_OS)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ base_frequency * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_bf_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
```

```

||
|| REML criterion at convergence: 19481.2
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -11.6252  -0.3573   0.0068   0.3789  10.5554
||
|| Random effects:
||  Groups      Name      Variance Std.Dev.
||  SubjID:chlabel (Intercept) 23.212   4.818
||  SubjID        (Intercept) 18.541   4.306
||  Residual                9.232   3.038
|| Number of obs: 3240, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    5.617e-01  5.722e-01  5.700e+01   0.982   0.3304
|| base_frequency1 1.893e-01  1.069e-01  1.618e+03   1.772   0.0767 .
|| OS              1.290e+00  6.399e-01  5.700e+01   2.016   0.0485 *
|| LP              4.677e-02  4.652e-01  5.700e+01   0.101   0.9203
|| base_frequency1:OS 1.947e-01  1.196e-01  1.618e+03   1.628   0.1036
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) bs_fr1 OS      LP
|| bas_frqncy1  0.000
|| OS            -0.046  0.000
|| LP            0.038  0.000  0.021
|| bs_frqn1:OS  0.000 -0.047  0.000  0.000

```

4.2 N400 WORD DATA: FAMILY SIZE and OS as FIXED EFFECTS; LP as COVARIATE

```

# BINNED BY FAMILY SIZE

m_w_N400_fs_OS <- lmer(
  value ~ family_size * OS + LP +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_fs_w, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_w_N400_fs_OS)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ family_size * OS + LP + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_w, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 15561.4
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -10.9068  -0.3881   0.0130   0.4078   6.1805
||
|| Random effects:
||  Groups      Name      Variance Std.Dev.
||  SubjID:chlabel (Intercept) 5.237   2.289
||  SubjID        (Intercept) 4.575   2.139
||  Residual                3.304   1.818
|| Number of obs: 3240, groups:  SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error      df t value Pr(>|t|)
|| (Intercept)    2.618e-01  2.842e-01  5.700e+01   0.921   0.3608
|| family_size1    4.518e-01  6.394e-02  1.618e+03   7.066 2.37e-12 ***
|| OS              6.771e-01  3.179e-01  5.700e+01   2.130   0.0375 *
|| LP              1.231e-03  2.311e-01  5.700e+01   0.005   0.9958
|| family_size1:OS 9.090e-02  7.155e-02  1.618e+03   1.270   0.2041
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||              (Intr) fmly_1 OS      LP
|| family_siz1  0.000
|| OS            -0.046  0.000
|| LP            0.038  0.000  0.021
|| fmly_sz1:OS  0.000 -0.047  0.000  0.000

```

5 N400 NONWORD DATA : SYMMETRICAL MODEL, INTER-ACTION WITH LP AND OS

For the nonword N400, does the symmetric model we just ran replace the separate LP and OS models, or do all three get reported?***

The symmetric model is the cleanest solution — it tests both dimensions in the same model and provides the strongest evidence for the dissociation. The separate models are now superseded.

5.1 N400 NONWORD DATA: Family Size

5.1.1 Symmetric Model

```

contrasts(erp_fs_nw$complexity) <- c(-0.5, 0.5)
contrasts(erp_fs_nw$family_size) <- c(-0.5, 0.5)

m_nw_N400_fs_symmetric <- lmer(
  value ~ complexity * family_size * LP +
    complexity * family_size * OS +
    (1 | SubjID) + (1 | SubjID:chlabel),
  data = subset(erp_fs_nw, time_window == "N400"),
  control = lmerControl(optimizer = "bobyqa")
)

summary(m_nw_N400_fs_symmetric)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: value ~ complexity * family_size * LP + complexity * family_size * OS + (1 | SubjID) + (1 | SubjID:chlabel)
|| Data: subset(erp_fs_nw, time_window == "N400")
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: 32974.7
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -14.0373  -0.4804  -0.0160   0.4598  14.1912
||
|| Random effects:
|| Groups          Name          Variance Std.Dev.
|| SubjID:chlabel (Intercept)  6.392    2.528
|| SubjID         (Intercept)  5.568    2.360
|| Residual                6.100    2.470
|| Number of obs: 6480, groups: SubjID:chlabel, 1620; SubjID, 60
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)   -0.45163    0.31311   57.00007  -1.442 0.154665
|| complexity1   -0.46942    0.06147  4851.00008  -7.636 2.68e-14 ***
|| family_size1   0.08473    0.06147  4851.00008   1.378 0.168178
|| LP              0.01139    0.25458   57.00006   0.045 0.964463
|| OS              0.38123    0.35019   57.00006   1.089 0.280904
|| complexity1:family_size1 -0.44594    0.12295  4851.00008  -3.627 0.000290 ***
|| complexity1:LP   0.25409    0.04998  4851.00008   5.084 3.84e-07 ***
|| family_size1:LP   0.12220    0.04998  4851.00008   2.445 0.014528 *
|| complexity1:OS   0.23998    0.06876  4851.00008   3.490 0.000487 ***
|| family_size1:OS   0.06566    0.06876  4851.00008   0.955 0.339609
|| complexity1:family_size1:LP 0.22934    0.09997  4851.00008   2.294 0.021819 *
|| complexity1:family_size1:OS 0.14265    0.13751  4851.00008   1.037 0.299619
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) cmplx1 fmly_1 LP      OS      cm1:_1 cm1:LP f_1:LP cm1:OS f_1:OS c1:_1:L
|| complexity1  0.000
|| family_size1 0.000 0.000
|| LP           0.038 0.000 0.000
|| OS          -0.046 0.000 0.000 0.021
|| cmplxty1:_1 0.000 0.000 0.000 0.000 0.000
|| cmplxty1:LP 0.000 0.038 0.000 0.000 0.000 0.000
|| fmly_sz1:LP 0.000 0.000 0.038 0.000 0.000 0.000 0.000
|| cmplxty1:OS 0.000 -0.046 0.000 0.000 0.000 0.000 0.000 0.021 0.000
|| fmly_sz1:OS 0.000 0.000 -0.046 0.000 0.000 0.000 0.000 0.000 0.021 0.000
|| cmpl1:_1:LP 0.000 0.000 0.000 0.000 0.000 0.038 0.000 0.000 0.000 0.000
|| cmpl1:_1:OS 0.000 0.000 0.000 0.000 0.000 -0.046 0.000 0.000 0.000 0.000 0.021
||
|| # Effect Sizes
eta_squared(m_nw_N400_fs_symmetric, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||

```

```

|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----
|| complexity | 0.01 | [0.01, 1.00]
|| family_size | 3.91e-04 | [0.00, 1.00]
|| LP | 3.51e-05 | [0.00, 1.00]
|| OS | 0.02 | [0.00, 1.00]
|| complexity:family_size | 2.70e-03 | [0.00, 1.00]
|| complexity:LP | 5.30e-03 | [0.00, 1.00]
|| family_size:LP | 1.23e-03 | [0.00, 1.00]
|| complexity:OS | 2.51e-03 | [0.00, 1.00]
|| family_size:OS | 1.88e-04 | [0.00, 1.00]
|| complexity:family_size:LP | 1.08e-03 | [0.00, 1.00]
|| complexity:family_size:OS | 2.22e-04 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

```

```

# R-squared
r2_nakagawa(m_nw_N400_fs_symmetric)

```

```

|| # R2 for Mixed Models
||
|| Conditional R2: 0.667
|| Marginal R2: 0.013

```

5.1.2 Means and Contrasts

```

emm_options(lmer.df = "satterthwaite")

range(erp_bf_nw$LP)

```

Complexity × Family Size x LP Interaction

```

|| [1] -2.674523 3.111265

range(erp_bf_nw$OS)

```

```

|| [1] -1.975820 1.926921
# LP three-way follow-up
emm_3way_LP <- emmeans(m_nw_N400_fs_symmetric,
  ~ complexity * family_size | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)),
  lmerTest.limit = 6480)

```

```

LP_3way_contrasts <- compact_3way_table(
  emm_3way_LP,
  by_var = LP,
  condition_prefix = "LP",
  factor1 = complexity,
  factor2 = family_size
)
as.tibble(LP_3way_contrasts) |> print(n = 50)

```

```

|| # A tibble: 45 x 9
|| Comparison Condition Estimate SE df lower.CL upper.CL t.ratio p.value
|| <chr> <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex Large LP = -1 -0.99 0.4 60.3 -1.79 -0.19 NA NA
|| 2 Complex Small LP = -1 -0.62 0.4 60.3 -1.42 0.18 NA NA
|| 3 Simple Large LP = -1 0.06 0.4 60.3 -0.74 0.86 NA NA
|| 4 Simple Small LP = -1 -0.24 0.4 60.3 -1.04 0.56 NA NA
|| 5 Int: Simple - Complex across Small - Large LP = -1 -0.67 0.16 4851 NA NA -4.31 0
|| 6 Small - Large LP = -1 | complexity = Complex 0.37 0.11 4851 NA NA 3.36 0.001
|| 7 Small - Large LP = -1 | complexity = Simple -0.3 0.11 4851 NA NA -2.73 0.006
|| 8 Simple - Complex LP = -1 | family_size = Large 1.05 0.11 4851 NA NA 9.55 0
|| 9 Simple - Complex LP = -1 | family_size = Small 0.38 0.11 4851 NA NA 3.45 0.001
|| 10 Complex Large LP = -2.5 -1.37 0.71 60.3 -2.79 0.04 NA NA
|| 11 Complex Small LP = -2.5 -0.65 0.71 60.3 -2.07 0.77 NA NA
|| 12 Simple Large LP = -2.5 0.23 0.71 60.3 -1.19 1.64 NA NA
|| 13 Simple Small LP = -2.5 -0.06 0.71 60.3 -1.48 1.35 NA NA
|| 14 Int: Simple - Complex across Small - Large LP = -2.5 -1.01 0.27 4851 NA NA -3.7 0
|| 15 Small - Large LP = -2.5 | complexity = Complex 0.72 0.19 4851 NA NA 3.74 0
|| 16 Small - Large LP = -2.5 | complexity = Simple -0.29 0.19 4851 NA NA -1.49 0.136
|| 17 Simple - Complex LP = -2.5 | family_size = Large 1.6 0.19 4851 NA NA 8.26 0
|| 18 Simple - Complex LP = -2.5 | family_size = Small 0.59 0.19 4851 NA NA 3.03 0.002
|| 19 Complex Large LP = 0 -0.73 0.32 60.3 -1.37 -0.1 NA NA
|| 20 Complex Small LP = 0 -0.6 0.32 60.3 -1.23 0.04 NA NA
|| 21 Simple Large LP = 0 -0.05 0.32 60.3 -0.69 0.58 NA NA
|| 22 Simple Small LP = 0 -0.36 0.32 60.3 -0.99 0.27 NA NA
|| 23 Int: Simple - Complex across Small - Large LP = 0 -0.44 0.12 4851 NA NA -3.58 0
|| 24 Small - Large LP = 0 | complexity = Complex 0.13 0.09 4851 NA NA 1.53 0.127
|| 25 Small - Large LP = 0 | complexity = Simple -0.31 0.09 4851 NA NA -3.54 0

```

```

|| 26 Simple - Complex          LP = 0 | family_size = Large      0.68 0.09 4851      NA      NA      7.82 0
|| 27 Simple - Complex          LP = 0 | family_size = Small     0.24 0.09 4851      NA      NA      2.76 0.006
|| 28 Complex Large             LP = 1                          -0.47 0.42 60.3    -1.31 0.36 NA      NA
|| 29 Complex Small            LP = 1                          -0.58 0.42 60.3    -1.41 0.25 NA      NA
|| 30 Simple Large              LP = 1                          -0.16 0.42 60.3    -1      0.67 NA      NA
|| 31 Simple Small              LP = 1                          -0.48 0.42 60.3    -1.31 0.35 NA      NA
|| 32 Int: Simple - Complex across Small - Large LP = 1          -0.21 0.16 4851      NA      NA     -1.31 0.192
|| 33 Small - Large             LP = 1 | complexity = Complex   -0.1 0.11 4851      NA      NA     -0.91 0.36
|| 34 Small - Large             LP = 1 | complexity = Simple   -0.31 0.11 4851      NA      NA     -2.76 0.006
|| 35 Simple - Complex          LP = 1 | family_size = Large     0.31 0.11 4851      NA      NA      2.72 0.007
|| 36 Simple - Complex          LP = 1 | family_size = Small     0.1 0.11 4851      NA      NA      0.88 0.381
|| 37 Complex Large             LP = 3                          0.04 0.85 60.3    -1.66 1.74 NA      NA
|| 38 Complex Small            LP = 3                         -0.54 0.85 60.3    -2.24 1.16 NA      NA
|| 39 Simple Large              LP = 3                         -0.39 0.85 60.3    -2.09 1.31 NA      NA
|| 40 Simple Small              LP = 3                         -0.72 0.85 60.3    -2.42 0.98 NA      NA
|| 41 Int: Simple - Complex across Small - Large LP = 3          0.25 0.33 4851      NA      NA      0.76 0.45
|| 42 Small - Large             LP = 3 | complexity = Complex   -0.58 0.23 4851      NA      NA     -2.49 0.013
|| 43 Small - Large             LP = 3 | complexity = Simple   -0.33 0.23 4851      NA      NA     -1.42 0.155
|| 44 Simple - Complex          LP = 3 | family_size = Large     -0.43 0.23 4851      NA      NA     -1.84 0.066
|| 45 Simple - Complex          LP = 3 | family_size = Small     -0.18 0.23 4851      NA      NA     -0.77 0.441

```

```

# OS three-way follow-up
emm_3way_OS <- emmeans(m_nw_N400_fs_symmetric,
  ~ family_size * complexity | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480)

```

```

OS_3way_contrasts <- compact_3way_table(
  emm_3way_OS,
  by_var = OS,
  condition_prefix = "OS",
  factor1 = complexity,
  factor2 = family_size
)
as.tibble(OS_3way_contrasts) |> print(n = 50)

```

```

|| # A tibble: 45 x 9
||   Comparison          Condition Estimate SE    df lower.CL upper.CL t.ratio p.value
||   <chr>              <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex Large      OS = -1      -1.34 0.49 60.3    -2.31 -0.36 NA      NA
|| 2 Complex Small      OS = -1      -1.05 0.49 60.3    -2.03 -0.08 NA      NA
|| 3 Simple Large        OS = -1      -0.32 0.49 60.3    -1.29 0.66 NA      NA
|| 4 Simple Small        OS = -1      -0.63 0.49 60.3    -1.6 0.35 NA      NA
|| 5 Int: Simple - Complex across Small - Large OS = -1      -0.6 0.19 4851      NA      NA     -3.18 0.001
|| 6 Small - Large      OS = -1 | complexity = Complex 0.29 0.13 4851      NA      NA      2.15 0.032
|| 7 Small - Large      OS = -1 | complexity = Simple -0.31 0.13 4851      NA      NA     -2.35 0.019
|| 8 Simple - Complex    OS = -1 | family_size = Large 1.02 0.13 4851      NA      NA      7.66 0
|| 9 Simple - Complex    OS = -1 | family_size = Small 0.42 0.13 4851      NA      NA      3.16 0.002
|| 10 Complex Large      OS = -1.9     -1.85 0.76 60.3    -3.37 -0.33 NA      NA
|| 11 Complex Small      OS = -1.9     -1.44 0.76 60.3    -2.96 0.08 NA      NA
|| 12 Simple Large        OS = -1.9     -0.55 0.76 60.3    -2.07 0.97 NA      NA
|| 13 Simple Small        OS = -1.9     -0.87 0.76 60.3    -2.39 0.65 NA      NA
|| 14 Int: Simple - Complex across Small - Large OS = -1.9     -0.73 0.29 4851      NA      NA     -2.48 0.013
|| 15 Small - Large      OS = -1.9 | complexity = Complex 0.41 0.21 4851      NA      NA      1.97 0.049
|| 16 Small - Large      OS = -1.9 | complexity = Simple -0.32 0.21 4851      NA      NA     -1.53 0.126
|| 17 Simple - Complex    OS = -1.9 | family_size = Large 1.3 0.21 4851      NA      NA      6.12 0
|| 18 Simple - Complex    OS = -1.9 | family_size = Small 0.57 0.21 4851      NA      NA      2.76 0.006
|| 19 Complex Large      OS = 0        -0.77 0.32 60.3    -1.4 -0.13 NA      NA
|| 20 Complex Small      OS = 0        -0.62 0.32 60.3    -1.25 0.02 NA      NA
|| 21 Simple Large        OS = 0        -0.06 0.32 60.3    -0.69 0.58 NA      NA
|| 22 Simple Small        OS = 0        -0.37 0.32 60.3    -1 0.27 NA      NA
|| 23 Int: Simple - Complex across Small - Large OS = 0        -0.46 0.12 4851      NA      NA     -3.72 0
|| 24 Small - Large      OS = 0 | complexity = Complex 0.15 0.09 4851      NA      NA      1.72 0.085
|| 25 Small - Large      OS = 0 | complexity = Simple -0.31 0.09 4851      NA      NA     -3.54 0
|| 26 Simple - Complex    OS = 0 | family_size = Large 0.71 0.09 4851      NA      NA      8.17 0
|| 27 Simple - Complex    OS = 0 | family_size = Small 0.25 0.09 4851      NA      NA      2.91 0.004
|| 28 Complex Large      OS = 1        -0.2 0.47 60.3    -1.13 0.73 NA      NA
|| 29 Complex Small      OS = 1       -0.19 0.47 60.3    -1.12 0.74 NA      NA
|| 30 Simple Large        OS = 1         0.2 0.47 60.3    -0.73 1.13 NA      NA
|| 31 Simple Small        OS = 1       -0.1 0.47 60.3    -1.03 0.83 NA      NA
|| 32 Int: Simple - Complex across Small - Large OS = 1       -0.31 0.18 4851      NA      NA     -1.75 0.081
|| 33 Small - Large      OS = 1 | complexity = Complex 0.01 0.13 4851      NA      NA      0.1 0.921
|| 34 Small - Large      OS = 1 | complexity = Simple -0.3 0.13 4851      NA      NA     -2.37 0.018
|| 35 Simple - Complex    OS = 1 | family_size = Large 0.4 0.13 4851      NA      NA      3.13 0.002
|| 36 Simple - Complex    OS = 1 | family_size = Small 0.08 0.13 4851      NA      NA      0.66 0.507
|| 37 Complex Large      OS = 1.9      0.31 0.73 60.3    -1.15 1.78 NA      NA
|| 38 Complex Small      OS = 1.9      0.2 0.73 60.3    -1.26 1.67 NA      NA
|| 39 Simple Large        OS = 1.9      0.43 0.73 60.3    -1.03 1.9 NA      NA
|| 40 Simple Small        OS = 1.9      0.14 0.73 60.3    -1.33 1.6 NA      NA
|| 41 Int: Simple - Complex across Small - Large OS = 1.9     -0.19 0.28 4851      NA      NA     -0.66 0.512
|| 42 Small - Large      OS = 1.9 | complexity = Complex -0.11 0.2 4851      NA      NA     -0.55 0.581

```

```

|| 43 Small - Large      OS = 1.9 | complexity = Simple      -0.3  0.2  4851      NA      NA      -1.48  0.139
|| 44 Simple - Complex   OS = 1.9 | family_size = Large      0.12  0.2  4851      NA      NA      0.59  0.554
|| 45 Simple - Complex   OS = 1.9 | family_size = Small     -0.07  0.2  4851      NA      NA      -0.34  0.737

```

```

emm_complexity_FS_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ complexity | family_size,
  lmerTest.limit = 6480)

complexityxfamily_size_table <- as.tibble(compact_emm_table(
  emm_complexity_FS_N400,
  by_var = family_size
)) >> print()

```

Complexity × Family Size interaction averaged across LP

```

|| # A tibble: 6 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex      Large      -0.74  0.32  60.3   -1.38  -0.11  NA      NA
|| 2 Simple      Large      -0.05  0.32  60.3   -0.68   0.59  NA      NA
|| 3 Simple - Complex Large      0.7    0.09 4851    NA      NA      8.03   0
|| 4 Complex      Small      -0.6    0.32  60.3   -1.23   0.03  NA      NA
|| 5 Simple      Small      -0.35  0.32  60.3   -0.99   0.28  NA      NA
|| 6 Simple - Complex Small      0.25  0.09 4851    NA      NA      2.84  0.005

```

```

emm_complexity_CMP_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ family_size | complexity,
  lmerTest.limit = 6480)

family_size_x_complexity_table <- as.tibble(compact_emm_table(
  emm_complexity_CMP_N400,
  by_var = complexity
)) >> print()

```

```

|| # A tibble: 6 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Large      Complex      -0.74  0.32  60.3   -1.38  -0.11  NA      NA
|| 2 Small      Complex      -0.6    0.32  60.3   -1.23   0.03  NA      NA
|| 3 Small - Large Complex      0.14  0.09 4851    NA      NA      1.66  0.097
|| 4 Large      Simple      -0.05  0.32  60.3   -0.68   0.59  NA      NA
|| 5 Small      Simple      -0.35  0.32  60.3   -0.99   0.28  NA      NA
|| 6 Small - Large Simple      -0.31  0.09 4851    NA      NA     -3.54  0

```

transmute() creates a new data frame containing only the specified computations. factor() is used to encode a vector as a factor

```

emm_complexity_LP_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ complexity | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)),
  lmerTest.limit = 6480
)

complexityxLP_table <- as.tibble(compact_emm_table(
  emm_complexity_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)) >> print(n = 20)

```

Complexity × LP interaction averaged over Family Size:

```

|| # A tibble: 15 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex      LP = -1      -0.8    0.4   58.1   -1.6  -0.01  NA      NA
|| 2 Simple      LP = -1      -0.09   0.4   58.1   -0.89  0.71  NA      NA
|| 3 Simple - Complex LP = -1      0.71  0.08 4851    NA      NA      9.19   0
|| 4 Complex      LP = -2.5    -1.01   0.7   58.1   -2.42  0.39  NA      NA
|| 5 Simple      LP = -2.5    0.08   0.7   58.1   -1.32  1.49  NA      NA
|| 6 Simple - Complex LP = -2.5    1.09  0.14 4851    NA      NA      7.99   0
|| 7 Complex      LP = 0      -0.67   0.31  58.1   -1.29 -0.04  NA      NA
|| 8 Simple      LP = 0      -0.21   0.31  58.1   -0.84  0.42  NA      NA
|| 9 Simple - Complex LP = 0      0.46  0.06 4851    NA      NA      7.48   0
|| 10 Complex     LP = 1      -0.53   0.41  58.1   -1.35  0.3   NA      NA
|| 11 Simple     LP = 1      -0.32   0.41  58.1   -1.15  0.5   NA      NA
|| 12 Simple - Complex LP = 1      0.21  0.08 4851    NA      NA      2.54  0.011
|| 13 Complex     LP = 3      -0.25   0.84  58.1   -1.93  1.43  NA      NA
|| 14 Simple     LP = 3      -0.55   0.84  58.1   -2.24  1.13  NA      NA

```

```
|| 15 Simple - Complex LP = 3      -0.3  0.16 4851      NA      NA      -1.84  0.065
```

```
emm_FS_LP_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ family_size | LP,
  at = list(LP = c(-2.5, -1, 0, 1, 3)),
  lmerTest.limit = 6480
)

family_size_x_LP_table <- as.tibble(compact_emm_table(
  emm_FS_LP_N400,
  by_var = LP,
  condition_prefix = "LP"
)) |> print()
```

FS × LP interaction averaged over Complexity:

```
|| # A tibble: 15 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Large          LP = -1      -0.46  0.4   58.1   -1.26  0.33  NA      NA
|| 2 Small          LP = -1      -0.43  0.4   58.1   -1.23  0.37  NA      NA
|| 3 Small - Large LP = -1       0.03  0.08 4851    NA     NA     0.45  0.655
|| 4 Large          LP = -2.5     -0.57  0.7   58.1   -1.98  0.83  NA      NA
|| 5 Small          LP = -2.5     -0.36  0.7   58.1   -1.76  1.05  NA      NA
|| 6 Small - Large LP = -2.5     0.22  0.14 4851    NA     NA     1.59  0.112
|| 7 Large          LP = 0        -0.39  0.31  58.1   -1.02  0.24  NA      NA
|| 8 Small          LP = 0        -0.48  0.31  58.1   -1.11  0.15  NA      NA
|| 9 Small - Large LP = 0        -0.09  0.06 4851    NA     NA    -1.42  0.154
|| 10 Large         LP = 1        -0.32  0.41  58.1   -1.15  0.51  NA      NA
|| 11 Small         LP = 1        -0.53  0.41  58.1   -1.36  0.3   NA      NA
|| 12 Small - Large LP = 1        -0.21  0.08 4851    NA     NA    -2.6   0.009
|| 13 Large         LP = 3        -0.17  0.84  58.1   -1.86  1.51  NA      NA
|| 14 Small         LP = 3        -0.63  0.84  58.1   -2.31  1.05  NA      NA
|| 15 Small - Large LP = 3        -0.45  0.16 4851    NA     NA    -2.76  0.006
```

```
emm_complexity_OS_N400 <- emmeans(
  m_nw_N400_fs_symmetric,
  ~ complexity | OS,
  at = list(OS = c(-1.9, -1, 0, 1, 1.9)),
  lmerTest.limit = 6480
)

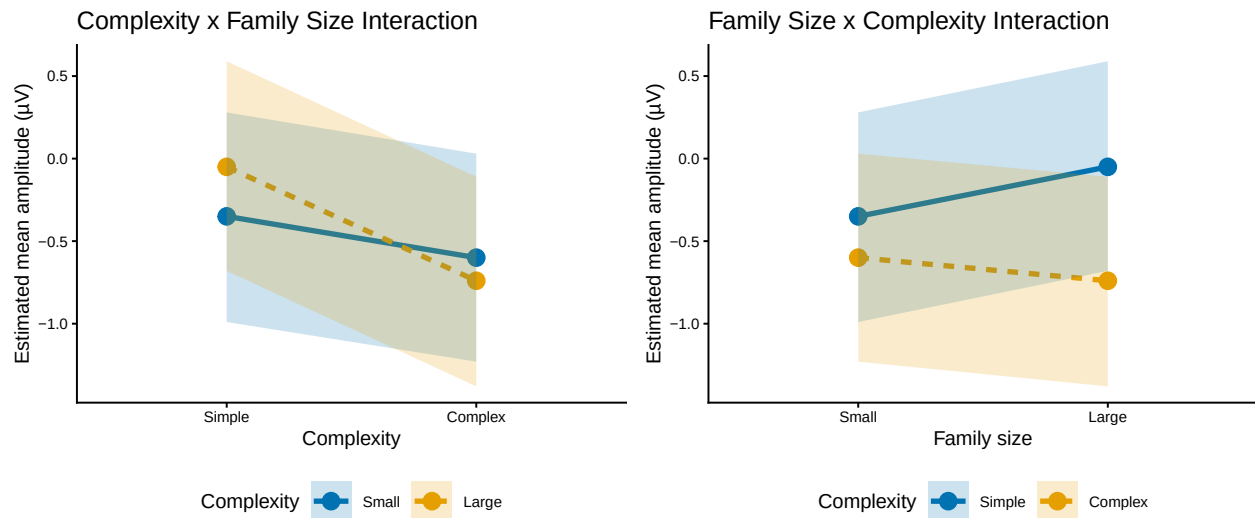
complexity_x_OS_table <- as.tibble(compact_emm_table(
  emm_complexity_OS_N400,
  by_var = OS,
  condition_prefix = "OS"
)) |> print(n = 20)
```

Complexity × OS interaction averaged over Family Size:

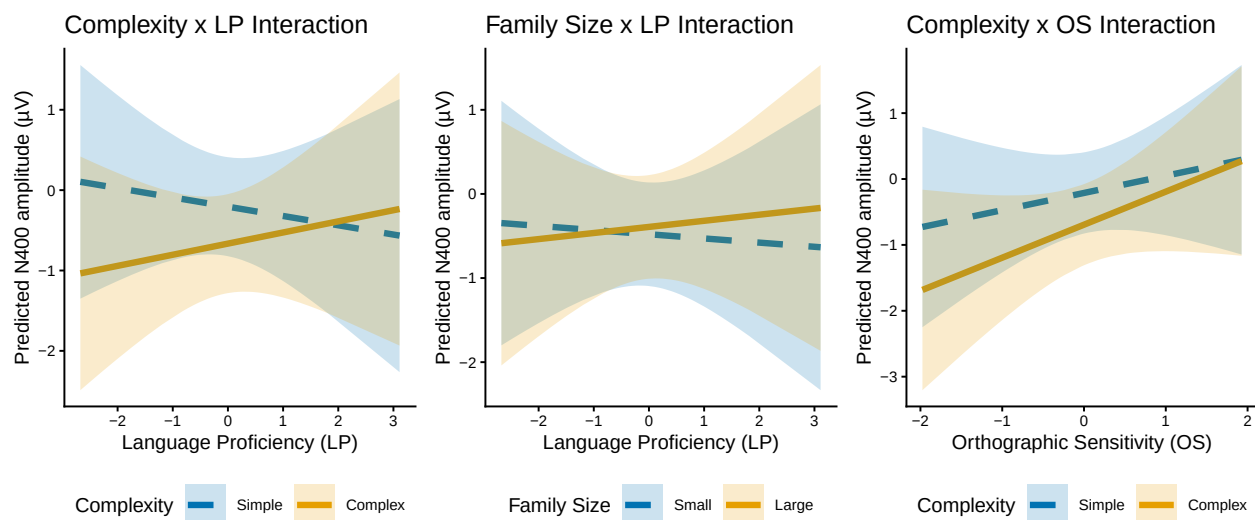
```
|| # A tibble: 15 x 9
||   Comparison      Condition Estimate      SE      df lower.CL upper.CL t.ratio p.value
||   <chr>          <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
|| 1 Complex          OS = -1      -1.19  0.48  58.1   -2.16 -0.23  NA      NA
|| 2 Simple           OS = -1      -0.47  0.48  58.1   -1.44  0.49  NA      NA
|| 3 Simple - Complex OS = -1       0.72  0.09 4851    NA     NA     7.65  0
|| 4 Complex          OS = -1.9     -1.65  0.75  58.1   -3.15 -0.14  NA      NA
|| 5 Simple           OS = -1.9     -0.71  0.75  58.1   -2.21  0.8   NA      NA
|| 6 Simple - Complex OS = -1.9     0.94  0.15 4851    NA     NA     6.38  0
|| 7 Complex          OS = 0        -0.69  0.31  58.1   -1.32 -0.06  NA      NA
|| 8 Simple           OS = 0        -0.21  0.31  58.1   -0.84  0.42  NA      NA
|| 9 Simple - Complex OS = 0         0.48  0.06 4851    NA     NA     7.84  0
|| 10 Complex         OS = 1        -0.19  0.46  58.1   -1.11  0.73  NA      NA
|| 11 Simple          OS = 1         0.05  0.46  58.1   -0.87  0.97  NA      NA
|| 12 Simple - Complex OS = 1         0.24  0.09 4851    NA     NA     2.68  0.007
|| 13 Complex         OS = 1.9     0.26  0.73  58.1   -1.19  1.71  NA      NA
|| 14 Simple          OS = 1.9     0.29  0.73  58.1   -1.17  1.74  NA      NA
|| 15 Simple - Complex OS = 1.9     0.03  0.14 4851    NA     NA     0.18  0.856
```

5.1.3 PLOT 2-WAY INTERACTIONS

FAMILY SIZE X COMPLEXITY Plot

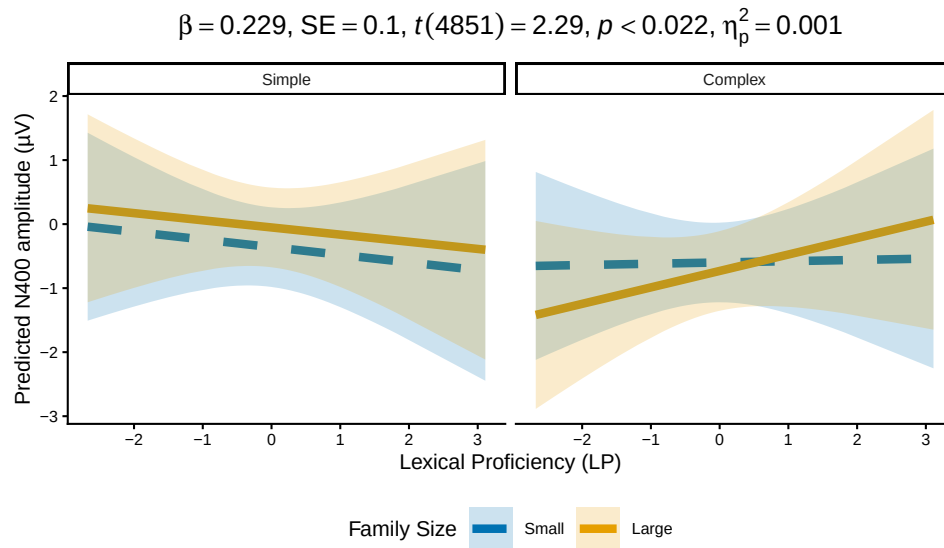


COMPLEXITY x LP Plot; FAMILY SIZE X LP Plot; COMPLEXITY x OS Plot



5.1.4 PLOT 3-WAY INTERACTIONS

Plots: COMPLEXITY x LP faceted by Family size; FAMILY SIZE x LP Faceted by Complexity

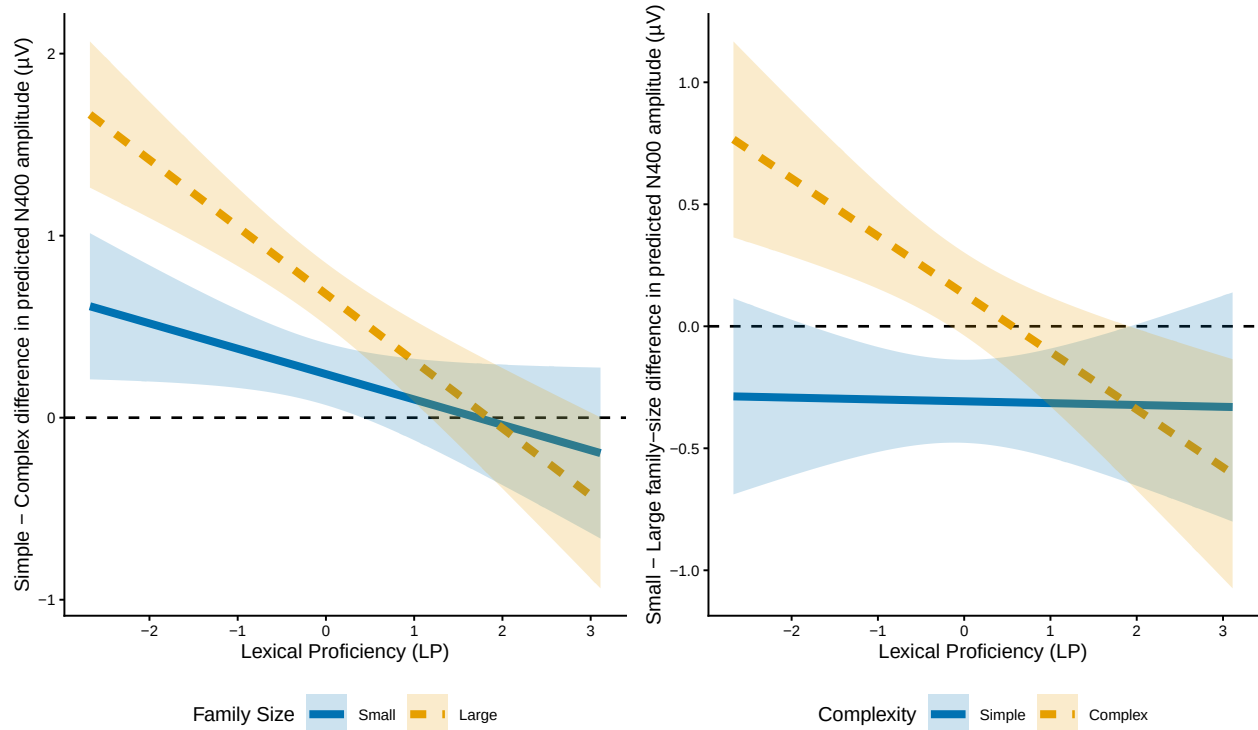


COMPLEXITY x LP and FAMILY SIZE x LP Interaction Plots Plot Simple-Complex difference across LP

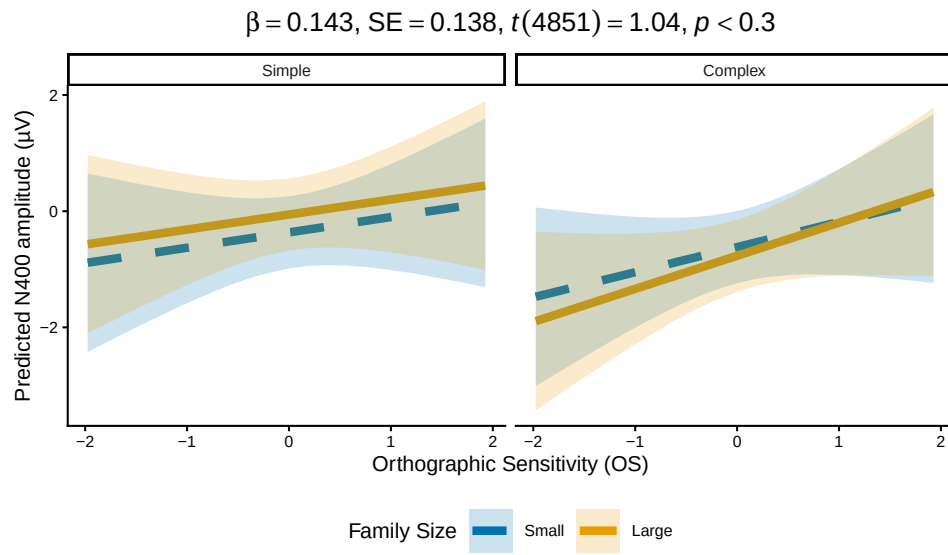
Plot Small Family - Large Family difference across LP

Print Both Plots

N400 NonWord Data



Plots: COMPLEXITY x OS faceted by Family size; FAMILY SIZE x OS Faceted by Complexity



Plot Small Family - Large Family difference across OS